

Figure 4: Conventional Risk Representation

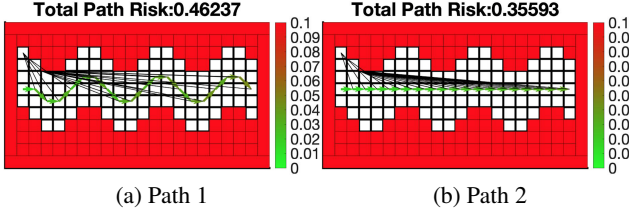


Figure 5: Proposed Risk Representation

closest obstacle and visibility, which are evaluated based on state alone. Their risk values at each state are combined using normalization and weighted sum (identical weights for both risk elements) and summed up along the entire path. Fig. 5b shows the result of the proposed probabilistic risk representation. All six risk elements from all three risk categories could be properly addressed. The risk at each state is now formulated as the probability of the robot not being able to finish the state, displayed in color. The probability of not being able to finish the path, as risk of the path, is computed using propositional logic and probability theory (Eqn. 3).

As shown in Fig. 4 and 5, the preference (lower risk) of the twisty path over straight path is switched when using the proposed risk representation instead of the conventional approach. This is because the proposed framework is able to consider more adverse effects due to action-dependent and traverse-dependent risk elements. It is worth to note that the fidelity of the model also depends on the accuracy of the failure probability value of each risk element at each state. How to derive those values more precisely, either using theoretical, empirical, or experimental methods, will be the focus of future work.

Another set of examples are shown in Fig. 6. Squeezing through the narrow passage between obstacles compromises distance to closest obstacle and visibility (locale-dependent risk elements), but optimizes action length, turn, tether length, and number of tether contacts. The path, which would be preferred by the conventional approach by only considering state-dependent risk elements (Fig. 6b), actually has a higher probabilistic risk value ($0.31 > 0.14$).

Conclusions

This paper proposes a formal definition of robot motion risk as the probability of the robot not being able to finish the path, which does not exist in the literature. This formal and general definition is applicable to any robotic agents loco-

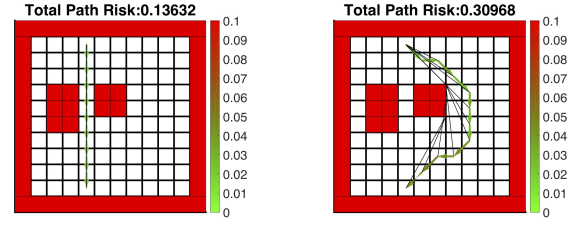


Figure 6: Path Risk Represented by Proposed Framework

moting in unstructured or confined environments. Therefore risk-aware motion is grounded as the motion with maximum probability of being safely finished. An explicit risk representation approach using propositional logic and probability theory is introduced. The use of these formal methods reveals that the risk the robot faces at each state along the path is not only a function of that current state itself, but also dependent on the traverse the robot took from the beginning. By considering not only locale-dependent, but also action-dependent and traverse-dependent risk elements, the proposed risk representation encompasses a comprehensive risk universe in unstructured or confined environments, e.g., motor overheat due to aggressive turning, sensor deterioration due to accumulated environmental interactions, etc., in addition to only obstacle-related risk. It also formally handles their dependencies on the history (longitudinal dependence). The inference using probability chain rule in the proposed risk framework also avoids the ill-founded additive assumption of risk along the entire path and other conservative relaxation techniques in the literature. With a simple lateral independence assumption and the formally reasoned longitudinal dependence, risk is computed as a single probability value of the robot not being able to finish the path. The proposed motion risk framework gives an explicit and intuitive comparison between different motion plans, or paths, for both human and robotic agents. It can be used as a metric to quantify safety for robust robot motion. Future work will focus on developing theoretical or experimental, other than empirical, methods for more precise individual risk values.

ACKNOWLEDGMENT

This work is supported by NSF 1637955, NRI: A Collaborative Visual Assistant for Robot Operations in Unstructured or Confined Environments.

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